

HIGH-CURRENT DC MOTOR CONTROLLER



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This controller system for industrial-grade high-current DC motors is based on ATmega8535 AVR microcontroller. It can be used in various applications such as conveyor control, shake testing, robotics and DC-operated drill machines.

Fig. 1 shows the block diagram of the AVR-based high-current DC motor controller. Two keys are provided to the user to control the rotation of the motor: One for selecting the clockwise direction and the other for selecting the anticlockwise direction of the motor. A 16x2 character LCD shows the clockwise and anticlockwise selection menu.

The direction of rotation of the motor is changed by reversing the polarity of the DC power supply to the motor. The polarity is reversed through four solidstate relays connected in H-bridge configuration.

ATmega8535 AVR microcontroller

The ATmega8535 is an 8-bit AVR microcontroller with 8 kB of in-system programmable flash. It has advanced RISC architecture with 130 powerful instructions, 32x8 general-purpose working registers and up to 16MIPS throughput at 16 MHz. By executing instructions in a single clock cycle, the ATmega8535 achieves throughputs approaching 1 MIPS per MHz allowing the system designer to optimise power consumption versus processing speed.

Solidstate relays

Solidstate relays (SSRs) function similar to electromechanical relays but

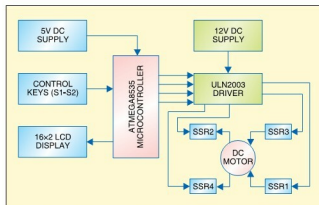


Fig. 1: Block diagram of AVR-based high-current DC motor controller



Fig. 2: Solidstate relay

these have no moving part to wear out and no contact bounce due to vibration. These are activated by AC or DC control signals from programmable logic controllers (PLCs), PCs, transistor-transistor logic (TTL) sources or microprocessor/microcontroller controls. There are three types of SSRs: photo-coupled, transformer-coupled and hybrid. Fig. 2 shows a typical solidstate relay.

Circuit description

Fig. 3 shows the circuit of the high-current DC motor controller system. The circuit operates off 12V DC supply available through CON1 connector. The 12V DC supply is fed to fixed-voltage regulator IC 7805 (IC1), motor-driver section comprising transistors T1 through T4, ULN2003 and solidstate relays. IC1 converts it into 5V DC, which is

then fed to the microcontroller and the LCD. Here, ATmega8535 microcontroller (IC2) is programmed for use as an interface between the motor and the user. As it has an internal oscillator, no external oscillator components are used in this circuit.

The 16-pin, 16x2 LCD module displays 'clockwise' or 'anticlockwise' message depending on the direction of rotation of the motor. Using variable resistance VR1 you can control the contrast of the LCD.

Port B of the microcontroller (IC2) is used to interface the data pins of the LCD. The RS and E pins of the LCD are

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V regulator
- IC2 - ATmega8535 AVR microcontroller
- IC3 - ULN2003 driver
- T1-T4 - BC547 npn transistor

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 220-ohm
- R2 - 10-kilo-ohm
- RNW1 - 4.7-kilo-ohm network resistor

Capacitors:

- C1 - 1000µF, 25V electrolytic
- C2 - 100µF, 16V electrolytic

Miscellaneous:

- SSR1-SSR4 - 24-200V DC load solidstate relay
- LCD1 - 16x2 LCD module
- S1-S2 - Push-to-on tactile switch
- 16-pin berg stick connector (male and female)
- 2-pin CPU connector (male and female)
- 8-pin CPU connector (male and female)
- DC motor